

What is your current role, what are you?

I am a developer.

And how long have you been working with software development?

Ah, software development, I have been working since 2018. More than 5 years.

And what AI tools do you use in your work?

Yes, recently the company, they launched an AI agent, which is called Klein. Klein is basically an AI agent, a copilot that helps you during development. It basically takes full control of your machine, it can, like, you ask it to solve some problem, right, that is happening in your code. Then it can understand the whole context of your project there, and can read the files, it basically assumes the role, your role, like that. It can send commands in the terminal, can write for you, can, anyway. And it does basically everything that you ask it there. So, yes.

And how often do you use this tool?

Ah, look, to tell you the truth, I am still a little resistant to using it. Sometimes I prefer to struggle with it myself, but I am trying to adapt my routine, right? Eh, there to solve some bug. So, like, it is from time to time, you know? It is not always that I use it, but when there is a problem there and I want a light, right? To understand, then I end up using, right, the agent resources, right, to try there to solve the problem, to give me at least an idea. From time to time.

And in what types of activity do you usually use AI?

Currently more development. I have also used it to, how can I say it, study code from already existing projects, right? Like, the project is very large, then you need there to summarize or at least understand some excerpt that you are not, you did not understand, right, while reading the code, so it is, understanding these projects that already exist, right? And development itself in general, bug fixes, development of new features, and I have also used it for unit tests, right? It generates unit tests based on your project.

And can you describe a recent case in which you used AI to perform some development activity?

Ah, today itself. I was going through a problem, there we are needing to use an implementation of a feature, right? The Wi-Fi Direct functionality. And the Wi-Fi Direct documentation, for the Android platform, it is a little, how can I say it, deprecated, right? It is very old. So, in some excerpts there, from the documentation, like, there are functions that do not even exist anymore, you know? So, it became a little difficult to develop this functionality and in some moments I needed to trigger the resource, like, man, it was giving a problem, because the way to develop a certain function call, is no longer supported. Then I had to trigger AI, right, to give me a light there and try to solve the problem, right? So, like, it basically managed to solve it there, make the correct function call and replacing information that was already deprecated in the documentation, in the library documentation itself. So, I believe it can search the internet, in online forums and can solve it, right? And some problems that sometimes the documentation itself cannot provide support for. So, basically that was the scenario, like, I was developing, there was a function that was no longer used, I called the AI, it managed to solve the problem. There in a few moments.

And how did you start this task? What did you receive, what did you need to do?

The process was actually very simple. As I said, I think it has access to the internet, so what I did was the following, I faced the problem, right? I saw that there was some compilation error there. Then I opened the project in the agent, right? It is basically a VS Code extension. So, for the project, it already has access to your entire project. And like, when you open a folder in VS Code, it has access to all the files in that folder. So, like, this also serves as input for it to understand the context. Then, basically, you ask what you want, like, I asked there, I, I do not remember now. It was for the connection part itself, right? B2B. So, I needed to make the connection between two

devices and using Wi-Fi Direct. And then I just opened the project in VS Code, using the extension, and asked. Look, can you implement the connection using Wi-Fi Direct. Then, I entered with a prompt and then it starts doing the thinking process. Basically it will first try to understand the context and establish some steps there. So, it will elaborate some hypotheses, right? Basically, this is basically already the agent working. Then it tries to say: "Look, I need to do this step, this step, this step, this step in your project in order to implement what you asked me, in this case it was the connection." Then it started suggesting, right, some changes in the project, then you need to allow, right, or reject, or you accept or reject, what it suggests. Then we go accepting there, right? First you take a look at what it suggested and you go saving it, right? So, as it goes suggesting, you can already go saving the changes that it presents to you. So, in the end, it, as it already goes, it had already been changing, right, during, it is an interactive process, you know? It asks if it can change, you accept, then it asks if it can change, you accept and while doing that it goes explaining, right, what it is thinking and what it is doing. Then it was basically that. In the end, I save all the files, compile and then it worked correctly.

Does your agent generate markdown?

It does. Because it also uses this to give context, to also explain the code, in the conversation history there itself, sometimes it has a styling of the text, right? So, it ends up being in markdown, the answer.

How many more or less does it generate in this process?

It depends a lot on the problem, you know? For example, it can, as I said, first try to establish some steps, you know? Some topics that it will try to do in order to solve the problem. So, for example, it can solve it in eight steps or it can solve it in 10 steps. Do you understand? So, all of this it goes explaining, right? And then, usually, each step is finalized with a code excerpt that it wants to change in your program, do you understand? But it varies a lot from problem to problem.

So it is not structured? Depending on your problem, it can act in different ways there.

That is it. It can suggest code, it can suggest running a command in the terminal. It is very versatile in that sense. It can at a certain moment, it does this a lot, it is, ah, I need to analyze such file.

Then it will read the file. Then it says: "Look, I found this function that may be useful." Then it goes there and changes another file, understood? So like it is a very interactive process and it is as if it were really a copilot, right? It is the proposal of this extension, of this AI. It is to be a copilot there in which it keeps helping you, right? And you keep approving or not. There is even a mode where you can let it solve the whole problem, like, you do not even need to give consent. I do not like this option very much, because it, sometimes, it does a lot of stupid things, right? But it is a possible option, you know?

It is an evolution of that tool we were using that was only for conversation, like a chat. It was an LLM, it was more of a chat. This one no, it really took control of what you want, of development.

So you keep confirming? Step by step, adapting. Can you do this, pause and adapt?

Yes, yes, you can, you can pause it at any moment, you can reject and it will try to find another solution or accept and then it will continue with its process, its reasoning.

But you cannot, like, it sent something. Then you say: "Ah, I think it can be this way." Will it do it or can you change it there by hand?

I can, I can, as I told you, pause, then like you can enter with a new prompt, you know?

But then is it all by prompt?

That is right, but like, it is aware of the whole context of the project. So, it can access everything in the project, literally everything.

And how do you validate these results?

It is, usually, as it is a process there during development, right? We validate, right, with tests, usually manual tests, but, obviously, you cannot completely discard the documentation, right? We, at least I, right? I do not trust AI 100%, I like to keep validating everything that it generates there. Sometimes I end up not accepting its solution completely, but I take part of the solution that it suggested, you know? Like, as it edits some excerpts there, there is an excerpt, for example, that I do not agree with, I go there, copy only what interests me and save it myself, like, manually, do you understand? And let it continue the process. So, basically, I take advantage of what I consider useful from the AI and together with the documentation. I do not trust AI completely, because sometimes it generates things that have nothing to do with it. So, you cannot completely trust the model.

Has the use of AI changed the way you work? If so, how?

Ah, yes, for sure. Mainly for these phases that we are going through now, which is the POC phase, right? Where we need to deliver results a little faster, right? So, for example, now we have a tight development cycle, we need to deliver these POCs by Tuesday and AI ends up being a big help, mainly to do repetitive things that we, like, already know. So, we would only waste time there, developing something that is laborious and that we could be spending time on something that would be a little more complex. It also ends up being useful to solve bugs, eventual bugs that we find during development. So, as it has a little more notion there, I do not say notion, I actually say, it has a little larger context, right? So it can sometimes find the solution to a bug. And it ends up increasing productivity, right, also during development.

And what difficulties do you encounter when using AI?

So, the main difficulty is that in reality it is. How can I say it? AI is not 100% intelligent, right? So, sometimes it ends up generating a lot of garbage. It is not only good things that it generates, right? Sometimes it generates a lot of things that have nothing to do with it. If you do not know, it was mainly there in the beginning, right? The configuration part of this agent was a little confusing, so, if you do not know correctly, it can end up doing stupid things on your computer. So, like, as it has access to the terminal, for example, it may even not know, like, it understands the prompts in the context of your project, right? The files of your project, the source code file and everything else. But, as it has access to your terminal, it can have access to your whole PC. Because I have used it at some moments and it like goes giving ls, for example, in the case of Linux. It keeps reading the whole directories there, understood? It keeps giving cd, keeps giving the directory, keeps exploring your whole PC. If you let it. So you have to be a little careful about this, it does not have these filters, understood? If you let it, it keeps changing everything. Even things that are critical in your operating system. So it is more these precautions like that and sometimes it does stupid things, so you cannot leave it alone working, without review.

So, you commented there that you had a little difficulty in the beginning configuring it, right? But how was your process of starting to use AI?

Ah, I had already used some tools that were a little simpler, with fewer functionalities, which was in the case of GitHub Copilot, at home. Then GitHub Copilot is a little simpler, because it, what happens? When you are developing there an excerpt of code, you put the comment, right, like as if it were a comment in the code and then underneath it tries to complete with what you asked. So, it is not really an agent, it is more of a little model, right, an LLM that can generate code. So, it was the first time that I started to use it, with this new model, from work, it was the first time that I started to use a real model, that assumes full control of your machine. So, it was a little, I say like, it was a somewhat interesting process to get used to, right? Like, what are the best prompts to put, how to configure it correctly for it to understand the project, basically configuring there, initial configuration.

And when you ask it to do something, what type of specification does it generate?

It says which steps it is going to do. At the beginning it already gives, I do not know, eight steps of what it intends to do there during, to solve that problem. So, in these processes that I said are interactive, what it generates are code excerpts or more critical commands there that it will need to use in the terminal. It is technical, very technical indeed. And then in the intervals, how can I say? It is, in what it needs to approve it is basically that. File changes and commands in your terminal. And then it explains, it keeps explaining why. It is as if it were a conversation, but a more interactive conversation. It is different from chat GPT, it is something more direct. You ask a question, it gives you an answer. You ask a question, it gives you an answer. With this agent, no. You enter with a prompt, it explains to you, then it asks for confirmation to change the file. Then if you accept, it says okay, now it will read the other file, it will try to understand the context there, then it asks for your confirmation and you save it. Understood? And so on. And that is how this interactive process really is.

And how do you feel using AI today? Do you prefer working with it or how it was before?

Look, I like AI to solve these cases like when we sometimes need a light, right? Then it can consult the internet, it ends up replacing a little of that research step, for example, it ends up being a little more convenient, for us, because it is basically you are going to ask and it will do everything for you. So, it will search on the internet, it will search in documentations and it will eventually end up finding some solution. I like it. I confess that I like it, right, it ends up, like, making you a little more productive, but at the same time I also like to understand what it did, line by line. I check line by line what it generated, I do not trust it completely. I also like to understand what it did, because basically that is what makes you a developer. You have to know how to solve the problem. If you leave it only for the AI to keep suggesting, there will come a time when you will have a bunch of code in your project that you do not know what anything does. You do not know, another detail too, sometimes it fails in the architectural solution, you know? It does not give you the best version of that code, sometimes it generates code that is redundant, that has code that does nothing, understood? So, like, even you, even if it solves the problem, sometimes it may not solve the problem in the best possible way. You need eventually to understand what it generated, you know? So, there is no way around it. And I prefer it like that, you know? I prefer that we need to keep reviewing and I think it is much more interesting than generating a bunch of code, because you do not know what it does and then it is even difficult to maintain. So, you have to be aware of that. But I like it, I like it. It increases productivity a lot for some activities.

And are there situations in which AI does not help as expected?

Look, I do not have much experience, for example, with refactoring. I do not like very much its style in architectural patterns. I do not think it is very good for that. Example, you are going to develop your POC there, fine, it took out a bunch of code there, nice. Then now you need to refactor your code. Then I think that, sometimes, it fails a lot, you know? Because it, as I said, generates a lot of redundant code, a lot of code that does nothing, so, there is a bunch of garbage in your code. A bunch of things in your project. If you let it, it leaves everything in a single file, a lot of lines there. So, I do not think this is cool and I still think that there is still a lot to evolve in this sense, of refactoring. It may be that there are other specific models for this, but this AI specifically I do not think it is good for refactoring or organizing your project, architecturally speaking.

What do you think is still missing to use AI better in development?

I have already worked with Android before, right? So, I have a little notion of Android development. So, some things I can filter and know what is right and what is wrong that it suggests to me. But I think that for someone who is starting, I think this can be a problem. For example, an AI that suggests something that you do not know and you are learning there, right? It can end up leading you to do something stupid, including in somewhat more serious scenarios, since, for example, this

AI, it can assume your terminal there, right, of your project, of your whole PC. So, you asked about recommendations, right? I think that, in this case, it would be to have theoretical knowledge of what you are doing, to have a base of what you are doing, before using a tool that can end up harming you more than helping you, you know? I think it is useful for those who know how to use it. It seems very basic, but I think you need to have a minimal notion of what you are doing, not only blindly delegate to it the role of developing and developing your work. You need to at least master it there and know what you are doing minimally.

For very large projects, it needs to understand a very absurd context. There is no way to put an AI to manage all of this, nor like always make the best possible code. It is still very dumb, sometimes, it is not intelligent at all. I do not think it will reach the level that it may end up replacing some stages there in the development process, but replacing development as a whole, I think it is very difficult. For vibe code it is interesting, understood? You are there, then you ask it to generate something, review it. I think that is the natural process.

And, there is also a point that is, like, AI, the new models, they tend, it is a possibility, to be worse than previous versions, because it ends up that the training of these more recent models. It includes content generated by AI itself, it keeps feeding itself back. So the tendency is to end up worsening the results compared to models that before were trained with real data, it was not generated by artificial intelligence. There will come a time when the curve will end up falling, the learning curve of these models, will end up falling instead of rising. Because, for example, there are already sites today that are no longer used as before, StackOverflow is around, Wikipedia, which was heavily used to train these models there, almost on the verge of bankruptcy, asking for donations not to close because people are no longer using them. People only want to use AI nowadays to search, they do not even use Google anymore, for example.